KALIMATI FOOD PRICE & INFLATION TRACKER

Data Analysis and Visualization — Project Documentation

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1. Introduction

Food prices are an important part of household expenditure and economic activity. Changes in the prices of vegetables, fruits, and other agricultural commodities can affect consumers, retailers, wholesalers, and businesses. Analyzing historical food-price data can help identify price patterns, compare commodities, observe market fluctuations, and communicate useful insights through data visualization.

The Kalimati Food Price & Inflation Tracker is an end-to-end data analytics project developed to analyze wholesale food-price data published by Kalimati Fruits and Vegetable Market, Nepal. The project collects historical price information, prepares it for analysis, stores structured records in a database, performs exploratory data analysis (EDA), calculates price indicators, and presents findings through visualizations and a Power BI dashboard.

The project demonstrates how raw data from a public website can be transformed into structured information that supports data-driven understanding of market-price patterns.

The primary focus of the project is descriptive and exploratory analysis. It examines the available data to understand how recorded prices vary across commodities and over time. The project also develops a selected-basket price index to summarize the price movements of a predefined group of commodities.

The selected-basket index is an analytical indicator created for this project. It is not Nepal's official Consumer Price Index (CPI), nor is it an official measurement of national inflation.

2. Project Objectives

The major objectives of the project are:

1. To collect historical wholesale food-price data from the Kalimati Market website.
2. To inspect the structure, size, data types, and overall quality of the collected dataset.
3. To identify missing values, duplicate records, invalid data types, inconsistent price values, and potentially unusual observations.
4. To clean and transform the data into a format suitable for analysis.
5. To use Python libraries such as Pandas and NumPy for data processing and statistical calculations.
6. To use SQLite and SQL for structured data storage and querying.
7. To perform exploratory data analysis to understand commodity-level and time-based price patterns.
8. To calculate a selected-basket price index and examine changes in selected commodity prices.
9. To create visualizations that communicate patterns and findings clearly.
10. To present key metrics and analytical results through a Power BI dashboard.
11. To document the analytical process and organize the project for reproducibility and future development.

3. Dataset Overview

3.1 Data Source

The data used in this project was collected from the official website of Kalimati Fruits and Vegetable Market, Nepal:

https://kalimatimarket.gov.np/price

The website publishes commodity-price information for different dates. The project uses the historical records available for the selected analysis period.

3.2 Dataset Coverage

The dataset covers the period from January 1, 2026, to July 31, 2026.

The dataset contains the following:

Dataset characteristic	Description
Data source	Kalimati Market website
Start date	January 1, 2026
End date	July 31, 2026
Total price records	21,095
Unique commodities	136
Overall mean recorded average price	Approximately Rs. 142.97
Primary analytical unit	Commodity price observation

Main purpose	Wholesale food-price analysis

The overall mean recorded average price of Rs. 142.97 is calculated across the available `avg` price observations. Since commodities can have different measurement units, this figure is a descriptive statistic for the dataset rather than the cost of a standardized basket of food.

3.3 Dataset Attributes

The primary dataset contains the following fields:

Column	Description
`date`	Date associated with the price observation
`commodity`	Name of the food or agricultural commodity
`unit`	Measurement unit associated with the commodity
`min`	Reported minimum price
`max`	Reported maximum price
`avg`	Reported average price

The date field supports time-based analysis. The commodity and unit fields support grouping and comparisons. The minimum, maximum, and average price fields provide numerical measures for descriptive statistics and visualization.

The dataset is structured as observational market-price data. Each record represents a published price observation rather than an individual consumer purchase.

3.4 Data Collection Process

The data collection process uses Python libraries, including Requests and BeautifulSoup, to retrieve and parse the relevant information from the Kalimati Market website.

The general workflow is as follows:

1. Identify the website page containing historical commodity prices.
2. Inspect how the website requests and returns price information for a selected date.
3. Send requests to retrieve the required data.
4. Parse the returned page content and extract relevant commodity-price fields.
5. Organize the extracted information into a structured tabular format.
6. Save the collected records for subsequent cleaning, storage, and analysis.

The website's date-selection mechanism and request format must be handled correctly to retrieve the intended historical records. The collected data is then processed into a format that can be used by Pandas and SQLite.

4. Tools and Technologies Used

4.1 Python

Python is the primary programming language used to collect, process, analyze, and visualize the dataset. Its data-analysis ecosystem supports the implementation of the project's analytical workflow.

4.2 Pandas

Pandas is used for loading tabular data, inspecting columns, handling dates, identifying missing values, grouping records, calculating descriptive statistics, and creating summary tables.

4.3 NumPy

NumPy supports numerical calculations and transformations, including calculations required for price indices and other analytical measures.

4.4 Requests and BeautifulSoup

Requests is used to retrieve web content, while BeautifulSoup is used to parse relevant HTML content and extract the required data fields.

4.5 SQLite

SQLite is used to store structured price records in a local database. This allows the data to be retrieved through SQL queries and used by Python-based analysis scripts.

4.6 PostgreSQL and pgAdmin

The data was also migrated from SQLite into PostgreSQL to enable additional SQL-based analysis using pgAdmin. This provides an opportunity to practice database querying, aggregation, data-quality checks, and analytical SQL techniques.

The PostgreSQL database is an additional analytical environment; the original SQLite database remains the project's local source database.

4.7 Matplotlib and Seaborn

Matplotlib and Seaborn are used to visualize price trends, commodity comparisons, and other patterns. These visualizations make numerical results easier to interpret.

4.8 Power BI

Power BI is used to present selected project findings through an interactive dashboard consisting of three pages:

1. Market Overview
2. Commodity Price Analysis
3. Seasonal and Market Insights

4.9 Jupyter Notebook, VS Code, Git and GitHub

Jupyter Notebook provides an interactive environment for executing analysis step by step. Visual Studio Code is used for development and project management. Git and GitHub support version control, code organization, and publication of the project.

5. Data Cleaning and Transformation Techniques

Data cleaning and transformation are essential because raw data collected from websites may contain inconsistent data types, missing observations, duplicate records, or values that require validation. Performing these checks before analysis improves the reliability and interpretability of the results.

The cleaning process is implemented using Python and Pandas.

5.1 Loading and Inspecting the Dataset

The first step is to load the price records into a Pandas DataFrame.

The dataset is inspected using methods such as:

- `head()` to view the initial records.
- `shape` to determine the number of rows and columns.
- `info()` to inspect column names, data types, and non-null counts.
- `describe()` to calculate descriptive statistics for numerical columns.
- `isnull().sum()` to count missing values in each column.

These checks provide an initial understanding of the dataset and help identify issues that may need further investigation.

5.2 Data Type Validation

The date field is converted into a proper datetime format so that records can be grouped by day and month, sorted chronologically, and used in time-series analysis.

The minimum, maximum, and average price fields are converted into numeric data types where necessary.

Converting these fields to appropriate data types is important because calculations such as averages, differences, and percentage changes require numerical values rather than text.

When conversion is performed using error handling, values that cannot be interpreted as numbers can be identified and investigated instead of silently being treated as valid prices.

5.3 Missing Value Analysis

Missing values are examined using Pandas' missing-value detection functions.

Missing information can affect descriptive statistics and trend analysis. For example, an absent average price prevents that observation from contributing to calculations involving `avg`.

The analysis checks missing values across the dataset, with particular attention to dates, commodity names, and numerical price fields.

Missing values should be handled according to their meaning. Records with missing optional fields may still contain useful information, while records with missing essential fields may be unsuitable for particular analyses.

The project does not assume that every missing value should automatically be replaced with zero. Zero is a genuine numerical value and should not be used as a substitute for an unknown price.

5.4 Duplicate Record Detection

Duplicate records are checked using Pandas' duplicate-detection functionality.

Duplicate observations can inflate record counts and influence aggregate statistics if the same price observation appears more than once.

The analysis examines exact duplicate rows and can also investigate repeated date–commodity–unit combinations.

Repeated combinations should be reviewed carefully because they do not necessarily represent errors. Multiple records may occasionally be legitimate, depending on the source data structure.

Therefore, duplicate detection is treated as a data-quality check, and records should only be removed when their duplication is confirmed.

5.5 Price Value Validation

The numerical price fields are examined for invalid or suspicious values.

The checks include:

- Negative minimum, maximum, or average prices.
- Average prices lower than reported minimum prices.
- Average prices higher than reported maximum prices.
- Missing or non-numeric price values.
- Unusually high or low price observations that may require investigation.

For records where minimum, average, and maximum prices are expected to follow their conventional ordering, the following relationship is checked:

Minimum price ≤ Average price ≤ Maximum price.

A violation may indicate a data-entry problem, a parsing issue, or an inconsistency in the source data. However, unusual values should be investigated before being classified as errors.

5.6 Date and Commodity Validation

The date field is checked to ensure that observations fall within the intended analysis period.

Commodity names are examined for missing values and inconsistencies. The analysis also checks whether the commodity names selected for the basket are present in the dataset.

Correct commodity identification is important because differences in spelling, formatting, or naming can cause the same commodity to be treated as separate categories.

The measurement unit is retained so that commodity comparisons can be interpreted correctly.

5.7 Potential Outlier Detection

Potential outliers can be investigated using the Interquartile Range (IQR) method.

The IQR is calculated as:

$$IQR = Q3 - Q1$$

where Q1 is the first quartile and Q3 is the third quartile.

The conventional lower and upper boundaries are:

$$\text{Lower boundary} = Q1 - 1.5 \times IQR$$

Upper boundary = Q3 + 1.5 × IQR

Observations outside these boundaries are flagged as potential outliers.

For market-price data, outlier analysis should be performed within meaningful groups, such as individual commodities, because prices for different commodities can have substantially different typical levels.

An unusually high price may reflect a genuine market movement rather than an error. For this reason, outlier detection is used to identify records for investigation rather than automatically remove them.

5.8 Data Transformation and Feature Engineering

After initial inspection and cleaning, the dataset is transformed to support additional analysis.

The main transformations include:

Date transformation: The date field is converted into a datetime representation to enable chronological sorting and date-based grouping.

Monthly features: The month number and month name are derived from the date field to support monthly aggregation and visualization.

Monthly commodity averages: Records are grouped by commodity and month to calculate average monthly prices.

Selected-basket aggregation: The average prices of a predefined set of commodities are combined to produce a daily basket-price measure when observations are available.

Basket index calculation: The basket price is normalized relative to its initial valid observation to create an index with a reference value of 100.

Commodity-level price indices: Monthly average prices for individual commodities are normalized to their first available monthly average to facilitate comparisons of relative price movements.

Price-change calculations: The first and last available monthly average prices are compared to calculate percentage changes for each commodity.

These transformations create additional analytical variables without changing the meaning of the original source-price fields.

5.9 Selected-Basket Price Index

The project defines a basket containing ten selected commodities:

- आलु रातो(लाम्चो)
- प्याज सुकेको (भारतीय)
- गोलभेडा ठूलो(नेपाली)
- काउली तराई
- बन्दा(लोकल)
- गाजर(तराई)
- काक्रो(लोकल)
- भिण्डी
- लौका
- मूला सेतो(लोकल)

For each date, the available average prices of the selected commodities are combined into a basket-price measure.

The index is calculated by comparing each day's basket price with the first valid basket price:

Basket Index = (Current Basket Price / Base Basket Price) × 100

An index value of 100 represents the base observation. An index above 100 indicates that the basket price is higher than its base value, while an index below 100 indicates that it is lower.

The number of available basket commodities is also monitored because the daily basket price may be based on fewer than ten commodities when some observations are unavailable.

Consequently, changes in commodity availability can affect the interpretation of the index. This indicator should not be confused with a formally weighted CPI or a complete measure of food inflation.

5.10 Commodity-Level Normalized Price Index

The project also calculates a normalized price index for individual commodities.

For each commodity, its first available monthly average price is used as the base:

Commodity Price Index = (Current Monthly Average Price / First Available Monthly Average Price) × 100

This allows commodities with different price levels to be compared in terms of relative changes rather than only their raw prices.

For example, a commodity with an index of 120 has a monthly average price approximately 20% higher than its own base-period average. This interpretation applies to that commodity's normalized series, not to the overall market.

The first available month may differ between commodities. Therefore, comparisons should consider the base month and data coverage.

6. Exploratory Data Analysis (EDA)

Exploratory Data Analysis is performed to understand the dataset's characteristics, identify patterns, and generate insights before drawing conclusions.

The analysis combines numerical summaries, grouped calculations, and visualizations.

6.1 Descriptive Statistical Analysis

Descriptive statistics are used to summarize the numerical price fields.

The analysis examines measures such as:

- Count of available observations.
- Mean price.
- Standard deviation.
- Minimum and maximum recorded values.
- Quartiles and median-related distribution information.

These statistics help explain the distribution and spread of recorded prices.

The overall mean recorded average price across the dataset is approximately Rs. 142.97. This statistic summarizes the recorded `avg` values, but it should be interpreted cautiously because different commodities may be measured in different units.

6.2 Commodity-Level Price Comparison

The dataset contains 136 unique commodities, allowing the analysis to examine differences in their recorded price levels.

Commodity-level summaries are calculated by grouping observations by commodity and calculating statistics such as mean, minimum, maximum, and observation count.

Among the recorded commodity-level averages, the following results were identified:

Measure	Commodity	Mean recorded average price
Highest mean price	सिताके च्याउ	Approximately Rs. 899.42

Lowest mean price मूला सेतो(लोकल) Approximately Rs. 15.54

These results demonstrate the substantial variation in recorded price levels across commodities.

However, this ranking does not necessarily indicate that one commodity is economically more expensive than another in a standardized sense. Different commodities may use different measurement units, and their average prices must be interpreted alongside those units.

6.3 Monthly Price Analysis

The dataset is grouped by month to calculate monthly average prices and summarize commodity coverage.

Monthly aggregation reduces the complexity of daily observations and makes it easier to compare broad price patterns over the study period.

The analysis also identifies the commodity with the highest monthly average recorded price in each month.

This helps examine whether the ranking of high-priced commodities remains consistent or changes over time.

Monthly averages should be interpreted alongside the number of observations available in each month. Changes in which commodities are reported can affect the composition of the overall average.

6.4 Daily Selected-Basket Price Trends

A daily basket-price measure is calculated from the selected ten commodities.

The daily basket-price series is visualized over time, and an index is calculated relative to the first valid basket-price observation.

This analysis provides a compact view of price movements among the selected commodities.

The basket index can help identify periods when the selected basket was relatively more expensive or less expensive than at the beginning of the observation period.

The number of available commodities is also tracked to help interpret daily changes. A shift in the index may reflect price movements, differences in commodity availability, or both.

6.5 Monthly Commodity Price Heatmap

A heatmap is used to compare monthly average prices across commodities.

The underlying data is organized into a matrix in which commodities form the rows and months form the columns. Each cell represents the monthly average recorded price for a particular commodity.

The visualization helps identify differences in price levels and changes across months.

Because the dataset contains many commodities, a smaller heatmap is used to improve readability. The selected subset focuses on commodities with higher mean prices in the available monthly summary.

The heatmap is intended for exploratory comparison rather than as a complete representation of all 136 commodities. Missing cells may indicate that a monthly observation is unavailable.

6.6 Monthly Trends for Selected Basket Commodities

Monthly average prices are plotted for the commodities included in the selected basket.

These line charts make it possible to examine whether the recorded prices of individual commodities increase, decrease, or fluctuate over the study period.

Comparing several commodities in the same chart can reveal that their price movements do not necessarily follow the same pattern.

The visualizations provide a more detailed view than the combined basket index because individual commodity movements can differ even when the overall basket price changes only slightly.

6.7 Normalized Commodity Price Trends

Normalized commodity indices are used to compare relative price changes across commodities with different base prices.

Each commodity's first available monthly average is assigned an index value of 100. Subsequent monthly averages are expressed relative to that baseline.

This approach helps distinguish relative changes from differences in absolute price levels.

For example, two commodities can have very different raw prices but similar normalized index movements if their prices change by similar percentages relative to their respective base periods.

The analysis also calculates percentage changes between each commodity's first and last available monthly average prices.

These calculations are useful for ranking observed price increases and decreases during the available period.

They do not establish why a price changed, and the first and last observations may not necessarily correspond to January and July for every commodity.

6.8 Data Quality Findings

The EDA process also examines data quality through checks for missing values, duplicate records, invalid price relationships, and potential outliers.

These checks help determine whether the dataset is suitable for the intended analysis and identify records requiring further inspection.

Data-quality findings should be interpreted according to the actual results of the notebook. The presence of a check does not automatically mean that an error was found, and a flagged outlier does not automatically mean the observation is incorrect.

The project uses these checks to improve transparency and to recognize potential limitations in the source data.

7. Key Findings

Based on the dataset overview and implemented analytical methods, the project supports the following findings.

7.1 Substantial Variation in Commodity Price Levels

The dataset contains 136 unique commodities, with mean recorded average prices ranging from approximately Rs. 15.54 for मूला सेतो(लोकल) to Rs. 899.42 for सिताके च्याउ.

This illustrates the substantial differences in recorded price levels across commodities. However, differences in measurement units must be considered before comparing raw prices.

7.2 Historical Price Data Supports Time-Based Analysis

The dataset covers seven months, from January through July 2026. This provides sufficient temporal coverage for exploratory daily and monthly analysis within the selected period.

The dataset can be grouped by date and month to examine price movements, compare monthly summaries, and identify commodities with notable changes.

However, the available period is not sufficient to establish year-over-year patterns or long-term seasonal behavior.

7.3 A Selected Basket Provides a Compact Summary of Price Movements

The selected-basket index summarizes the combined price behavior of ten predefined commodities relative to a base observation.

It provides a simple way to visualize whether the selected basket's recorded price is higher or lower than its base-period level.

Its interpretation depends on the selected commodities, their measurement units, and their availability on each date. It is therefore a project-specific analytical indicator rather than an official inflation measure.

7.4 Normalization Improves Relative Price Comparisons

Normalizing each commodity's price series to a base value of 100 makes it easier to compare relative price movements.

This approach helps distinguish commodities with large raw prices from commodities experiencing large proportional changes.

The first and last available monthly averages can also be used to identify commodities with the largest observed percentage increases and decreases.

7.5 Visualization Improves Interpretability

The project's line charts, monthly summaries, heatmap, and Power BI dashboard provide complementary views of the dataset.

Numerical summaries explain the magnitude of recorded prices, while visualizations help reveal changes over time and differences among commodities.

The three-page Power BI dashboard organizes the results into a market overview, commodity-level comparisons, and broader monthly or market insights.

8. SQL-Based Analysis

The project stores price records in SQLite and also uses PostgreSQL for additional SQL-based analysis.

SQL supports operations such as filtering, grouping, aggregation, sorting, and date-based analysis. PostgreSQL additionally provides analytical functions such as `LAG()` and `ROW_NUMBER()`.

The SQL analysis can be used to:

- Verify the total number of records and unique commodities.
- Calculate monthly summary statistics.
- Rank commodities by mean recorded price.
- Identify missing values and inconsistent price records.
- Detect repeated date–commodity–unit combinations.
- Compare monthly average prices.
- Calculate month-over-month percentage changes.
- Rank commodities by first-to-last available monthly price changes.

These queries complement the Python analysis by demonstrating how structured data can be analyzed directly through SQL.

The database migration also illustrates how a dataset can be moved from a lightweight local database into PostgreSQL for additional querying.

9. Power BI Dashboard

The Power BI dashboard presents selected results in a visual and accessible format.

9.1 Market Overview

The Market Overview page presents high-level market metrics and selected price trends. It provides an initial summary of the available dataset and its recorded price behavior.

9.2 Commodity Price Analysis

The Commodity Price Analysis page focuses on comparing commodity prices. It helps users examine differences in recorded price levels and identify commodities of interest.

9.3 Seasonal and Market Insights

The Seasonal and Market Insights page focuses on monthly patterns and broader observations from the available historical data.

Because the dataset covers only January through July 2026, these results should be described as observed monthly patterns rather than conclusive evidence of recurring seasonal effects.

The dashboard complements the Python notebook by allowing users to explore and communicate the findings visually.

10. Limitations of the Analysis

Several limitations should be considered when interpreting the results.

Limited time period: The dataset covers January 1 through July 31, 2026. It cannot establish year-over-year changes or long-term seasonal patterns.

Wholesale rather than retail prices: The data represents prices published by Kalimati Market. It may differ from retail prices, prices in other locations, and the actual amounts paid by consumers.

Different measurement units: Commodities may be measured in different units. Raw prices across different units are not always directly comparable.

Selected basket: The basket index is based on ten selected commodities and does not represent all food commodities or household expenditure patterns.

Availability of observations: Some commodities may not have records for every date or month. Differences in data availability can influence averages, indices, and comparisons.

Potential outliers: Unusually high or low prices may represent genuine market conditions or data-quality problems. Automated outlier detection cannot determine the cause on its own.

No causal analysis: The project identifies and describes recorded price patterns but does not establish the causes of price changes.

Not an official inflation measure: The selected-basket index is a project-specific indicator. It does not replace Nepal's official CPI or other official inflation statistics.

11. Conclusion

The Kalimati Food Price & Inflation Tracker demonstrates an end-to-end data analytics workflow using publicly available wholesale food-price data from Kalimati Market, Nepal.

The project begins with data collection and proceeds through data inspection, cleaning, transformation, database storage, exploratory data analysis, SQL querying, visualization, and dashboard development.

Pandas and NumPy support data manipulation and numerical calculations, while SQLite and PostgreSQL provide environments for structured data storage and SQL-based analysis. Matplotlib and Seaborn support exploratory visualization, and Power BI presents selected findings through an interactive dashboard.

The analysis describes the variation in recorded commodity prices, summarizes monthly patterns, and uses a selected-basket index and normalized commodity indices to examine relative price movements.

The project also demonstrates the importance of data-quality checks and careful interpretation. Missing observations, different measurement units, and the limited time period can affect the conclusions drawn from the data.

Overall, the project provides practical experience in data collection, data cleaning, exploratory data analysis, SQL, data visualization, and analytical reporting. It also creates a foundation for future extensions, including automated data collection, longer historical datasets, improved data validation, and more advanced market analysis.

12. Future Improvements

The project can be improved in several ways:

1. Automate data collection to retrieve new market-price observations regularly.
2. Extend the dataset to multiple years to support year-over-year analysis.
3. Improve data-quality monitoring and maintain a record of detected issues.
4. Develop a clearly documented and consistently defined basket-index methodology.
5. Add month-over-month and year-over-year comparisons when sufficient historical data is available.
6. Explore relationships between food prices and external variables such as rainfall, fuel prices, or other relevant economic indicators.
7. Expand the dashboard with additional filters and analytical views.
8. Develop a web-based interface for exploring historical price trends.
9. Improve reproducibility by documenting database creation, data loading, and analysis execution steps.

13. References

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End of Project Documentation